

STEEL AND CONCRETE • STRUCTURAL MODELING IN AUTODESK REVIT

REVIT FOR STRUCTURES

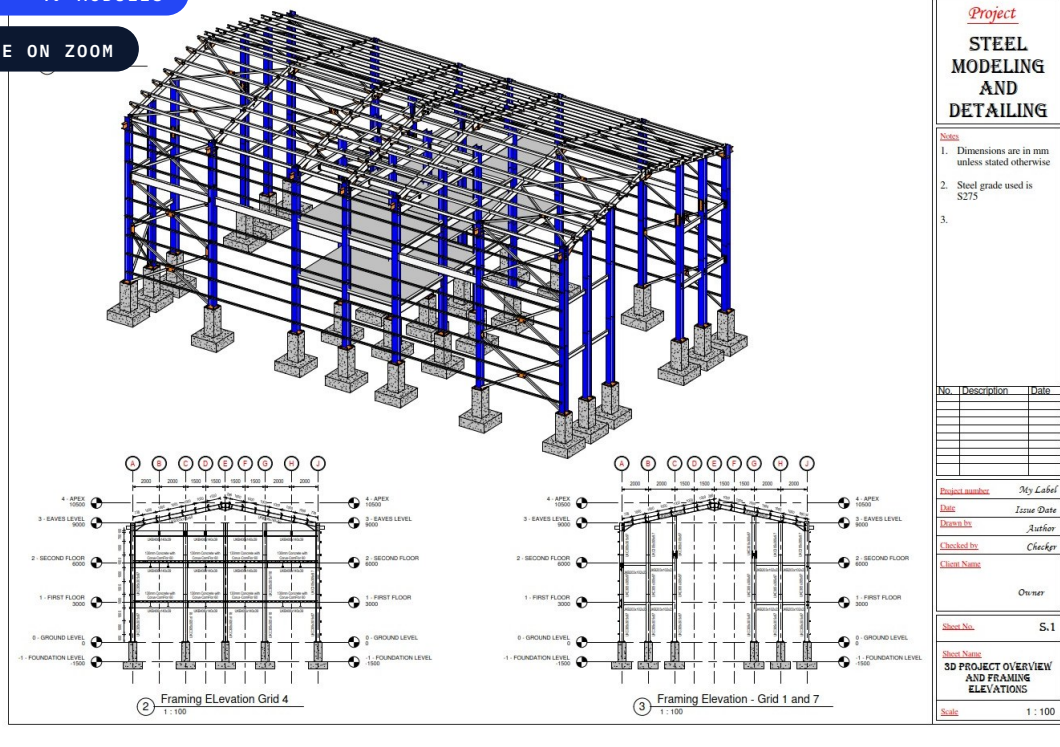
Build real, buildable structural models. Not just pretty renders.

2026

SEPT COHORT

2 TRACKS • 49 MODULES

100% LIVE ON ZOOM



STEEL PORTAL FRAME STRUCTURE • 3D PROJECT OVERVIEW AND FRAMING ELEVATIONS

FIG. 01

START

14 SEPT 2026

FORMAT

LIVE, ONLINE

RHYTHM

WEEKDAY EVENINGS

FEE

KSH 2,500 / TRACK

01 PROGRAM

p. 2

02 CONCRETE

p. 3

03 STEEL

p. 5

04 JOIN

p. 8

01 · THE PROGRAM

One program. Two buildable tracks.

Revit for Structures is a hands on Autodesk Revit program for engineering students and early career professionals who want to build real, buildable structural models. Two focused tracks, Reinforced Concrete and Steel, each run from first launch to finished documentation: modelling, reinforcement and connections, schedules, and sheets that print clean.

01

Model the whole frame

Columns to slabs, rafters to bracing.

02

Detail like a pro

Rebar, tags, bolts, plates and cuts.

03

Quantify & schedule

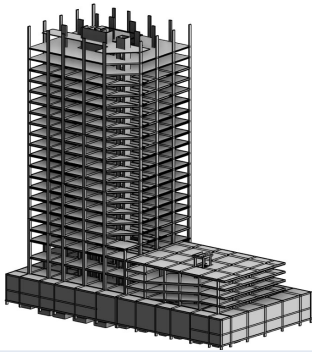
Bar bending schedules and takeoffs.

04

Document & deliver

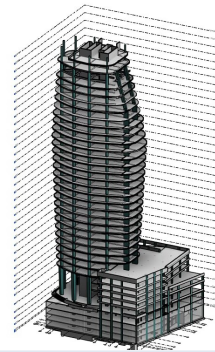
Sheets, PDFs, CAD and analysis handoff.

WHAT YOU WILL MODEL · REAL, STUDENT GRADE OUTPUT



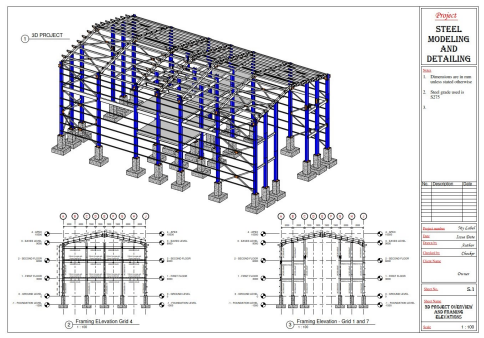
21 FLOORS + 2 BASEMENTS

FIG. 02



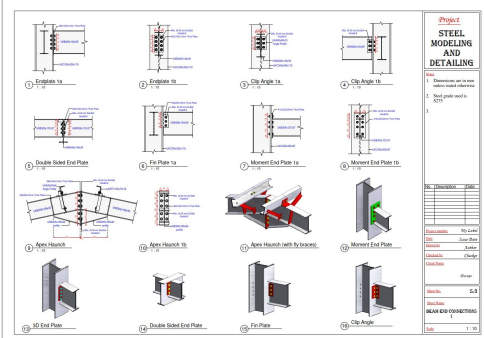
32 FLOORS + 4 BASEMENTS

FIG. 03



STEEL PORTAL FRAME STRUCTURE

FIG. 04



STEEL CONNECTIONS

FIG. 05

CHOOSE YOUR TRACK · OR TAKE BOTH

TRACK 01

REINFORCED CONCRETE

20 modules · Grids to BBS · p. 3

TRACK 02

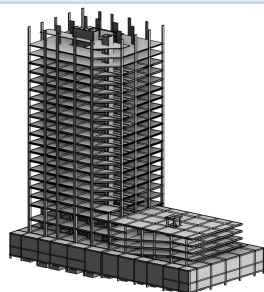
STEEL STRUCTURES

29 modules · Columns to connections · p. 5

02 · TRACK 01 · REINFORCED CONCRETE · 20 MODULES

FROM DATUMS TO DOCUMENTATION

Reinforced concrete, modelled to the rebar.



21 FLOORS + 2 BASEMENTS · REINFORCED CONCRETE TOWER FRAME, MODELLED IN REVIT

FIG. 02

A · GROUNDWORK · 01 TO 05

01 BRIEF OVERVIEW OF REVIT

How Revit thinks · Revit vs AutoCAD · Why BIM wins

02 SETTING UP THE SOFTWARE

Essential libraries · Autodesk extensions

03 BEGINNING WITH REVIT

New projects · Templates · Palettes · tabs · panels

04 DATUM ELEMENTS

Levels · Grids · Propagating extents · Grid bubble styling · Linking CAD for grids

05 3D ELEMENTS, FIRST PASS

Foundations · Columns · Beams · Slabs

B · MODELLING THE FRAME · 06 TO 10

06 COLUMNS

Placement · Loading families

07 FOUNDATIONS

Hosting on columns · Strip footings · Eccentricity · Basement slabs · Family libraries

08 BEAMS & SLABS

Copy level to level · Slab creation · Hollow pot & waffle

09 CURVED BEAMS

Modelling procedure, end to end

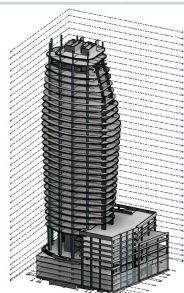
10 BEAM SYSTEMS & JUSTIFICATION

Beam system modelling · Layout rules · Justification rules · Deselecting cleanly

02 · TRACK 01 · REINFORCED CONCRETE, CONTINUED

Detail, reinforce, then deliver.

Stairs, rebar, tags, sheets.



32 FLOORS + 4 BASEMENTS · CURVED TOWER WITH LEVELS AND GRIDS, MODELLED IN REVIT

FIG. 03

C · DETAIL & REINFORCE · 11 TO 16

11 STAIRCASES & RAMPS

Stair types · Architectural vs structural ramps · Stairs modelled as ramps · Shaft openings

12 VISIBILITY, GRAPHICS & FILTERS

Graphic overrides, methods 1 to 4

13 ROOFS & TRUSSES

Sloped & gable ended roofs · System trusses · Custom trusses

14 REINFORCEMENT

Sections first · Foundations · Columns · Beams · Slabs · Stairs · Ramps

15 ANNOTATIONS

Dimensions · Spot elevations · Spot coordinates · Spot slopes

16 SECTION DETAILING

Breaklines · Hatch regions · Rebar tags · Comments · Element tags

D · DOCUMENT & DELIVER · 17 TO 20

17 SCHEDULES & TAKEOFFS

Bar bending schedules · Shape images in BBS · Material takeoffs · Sorting & filtering

18 DRAWINGS ON SHEETS

Titleblock sizes · System blocks · Custom blocks · Text & fields · Saving family templates · Duplicates · Placing views

19 PRINTING TO PDF

Sheet setup · Single sheets · Batch printing

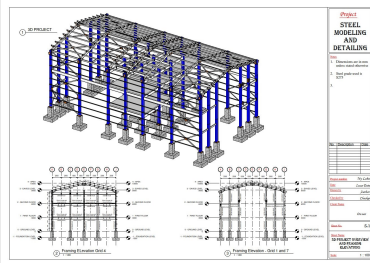
20 EXPORT & HANDOVER

Export options · To AutoCAD · To analysis

03 • TRACK 02 • STEEL STRUCTURES • 29 MODULES

COLUMNS • RAFTERS • BEAMS • BRACING

Steel, from columns to connections.



STEEL PORTAL FRAME STRUCTURE • EAVES, APEX AND FRAMING ELEVATIONS, MODELLED IN REVIT

FIG. 04

A • GROUNDWORK • 01 TO 04

01 BRIEF OVERVIEW OF REVIT

How Revit thinks • Revit vs CAD • Why BIM wins

02 SETTING UP THE SOFTWARE

Essential libraries • Autodesk extensions

03 BEGINNING WITH REVIT

New projects • Templates • Palettes • tabs • panels

04 DATUM ELEMENTS

Levels • Grids • Elevation markers • Propagating extents • Grid bubble styling • Linking CAD

B • THE STEEL FRAME • 05 TO 10

05 STEEL COLUMNS

Placement • Loading families

06 CONCRETE COLUMNS

Placement • Loading families

07 FOUNDATIONS

Hosting on columns • Extended family libraries

08 RAFTERS

Placement • Steel beam sections • Attaching columns • Copying grid to grid

09 STEEL BEAMS

Sections • Floor beams • Justification rules • Copying • Eaves level beams

10 BEAM SYSTEMS & JUSTIFICATION

Beam system modelling • Layout rules • Justification rules • Deselecting

Roofs, openings, bracing and views.

Composite decks to live sections.

C · ROOF, BRACING & VIEWS · 11 TO 21

11 COMPOSITE BEAM & SLAB

Composite beams · Composite slabs · Cantilevered edges

12 FLOOR OPENINGS

The shaft opening tool

13 CURVED BEAMS

Modelling procedure, end to end

14 SLOPED / SLANTED BEAMS

The offset method · Framing elevations

15 PURLINS & SIDE RAILS

Reference plane method · Beam system method · Side rails

16 BRACINGS

Vertical bracing · Horizontal bracing

17 VISIBILITY, GRAPHICS & FILTERS

Graphic overrides, methods 1 to 4 · Hiding elements

18 SECTIONS

Section types · Cutting live sections

19 REINFORCEMENT

Cover setup · Foundations · Columns · Slabs

20 SPLITTING ELEMENTS

Clean splits, buildable parts

21 CONNECTIONS, FIRST PRINCIPLES

Uploading connections · Applying them · Propagating across the frame

NEXT · THE CONNECTION LIBRARY

Thirteen connection families plus your own customs. Details on page 07.

SEE P. 07

The connection library.

Applied, propagated, detailed.

22 • CONNECTION TYPES • THE LIBRARY

Main types first, then every family you will meet on a real frame:

Main types

Haunch

Apex

Base plate

Gusset bracing

Gusset column base

Purlin / side rail

End plate, single sided (simple)

End plate, single sided (moment)

End plate, double sided (simple)

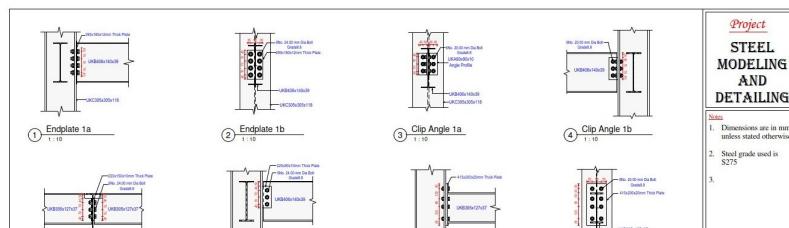
Fin plate

Clip angle

Column splice

Custom connections

APPLIED LIVE • MODULE 21 • UPLOAD • APPLY • PROPAGATE



STEEL CONNECTIONS • END PLATES, CLIP ANGLES AND MOMENT DETAILS AT 1:10

FIG. 05

D • FABRICATION • 23

23 FABRICATION & PARAMETRIC CUTS

Elements & modifiers • Where cuts live • Use & location

E • DOCUMENT & DELIVER • 24 TO 29

24 ANNOTATIONS

Dimensions • Spot elevations • Coordinates • Slopes • Custom bolt & plate tags

25 SECTION DETAILING

Breaklines • Hatch regions • Rebar tags • Comments • Element tags

26 SCHEDULES & TAKEOFFS

Bar bending schedules • Shape images in BBS • Material takeoffs • Sorting & filtering

27 DRAWINGS ON SHEETS

Titleblock sizes • System blocks • Custom blocks • Text & fields • Saving family templates • Duplicates • Placing views

28 PRINTING TO PDF

Sheet setup • Single sheets • Batch printing

29 EXPORT & HANDOVER

Export options • To AutoCAD • To analysis

Join the September cohort.

Live, online and hands on, from the third week of September.

SCHEDULE

START	Mon 14 Sept 2026
RHYTHM	Weekday evenings
FORMAT	Live on Zoom
TRACK	4 weeks each

Take either track, or both, back to back.

INVESTMENT

KSH **2,500**

per track • live instruction + materials

Students and early career engineers

No prior Revit experience needed.
Just a laptop that runs Revit.

HOW IT WORKS

1 Reserve your seat

Write to us via the website and pick a track.

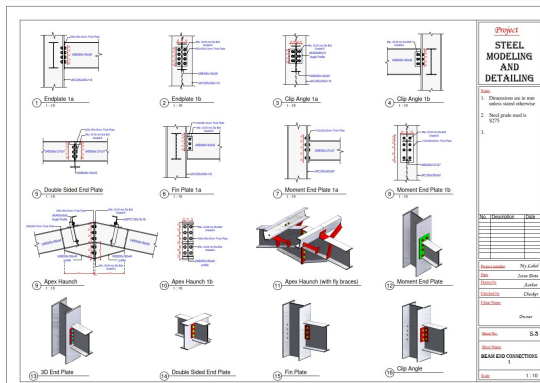
2 Set up Revit

We guide libraries and extensions before day one.

3 Build live

Evenings on Zoom. Model, detail, document.

GRADUATE WITH SHEETS LIKE THIS



STEEL CONNECTIONS • BEAM END CONNECTIONS (S.3) AT 1:10, MODELLED AND DETAILED IN REVIT

S.3

SEATS ARE LIMITED • SEPTEMBER COHORT

structratraining.co.ke

Reserve your track. Concrete, Steel, or both.

ENROL →

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